

# Graduation Project Progress Report 1

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| <b>Title</b>                      | Recommender systems using generative retrieval based on language models                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Summary<br/>(in 150 words)</b> | <p>We validated our core hypotheses across three independent public domains—Amazon Clothing, MovieLens 25M, and Goodreads Children—demonstrating that post-consumption category switching and attribute retention are universal phenomena, not dataset-specific artifacts. Users switch categories 56–70% of the time after a purchase or consumption event, yet retain key attributes (brand, author, series) at 3.76× to 16.39× above random chance during these transitions. Notably, Goodreads exhibited the strongest signal (author retention 16.39× random), and this held even after controlling for series-sequel confounding (series effect share: 0.00005%). A four-class user taxonomy—Author-Bridged Explorer, Discoverer, Author-Loyal Stable, and Genre-Stable—emerged with balanced distribution (18.7–31.3%), directly justifying our proposed intent classification. Temporal analysis confirmed that transitions are event-triggered, with switching rates monotonically increasing over longer intervals. These cross-domain findings strengthen our proposal's differentiation against single-dataset approaches and preemptively address reviewer concerns about robustness and confounding.</p> |
| <b>Advisor</b>                    | 김현준                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Period</b>                     | 2026. 03. 01 ~ 2026. 12. 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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# 1. Progress to Date

## 1.1. EDA Re-execution

After the Midterm Proposal, we re-conducted the EDA to address multiple issues identified in the initial analysis and to more rigorously validate the feasibility of our hypothesis.

### 1.1.1. The Initial EDA Problem

The initial EDA (`amazon_beauty_eda.ipynb`) contained the following problems.

First, there was a mismatch between the dataset and the title. The analysis was titled "Amazon Beauty," yet the data actually used belonged to the Clothing, Shoes and Jewelry category, causing fundamental confusion regarding the analysis target.

Second, the category extraction method was flawed. The lowest level (`cats[-1]`) of Amazon's hierarchical category metadata was used, which resulted in product attributes such as "Machine Washable" and "Padded footbed" being incorrectly classified as categories.

Third, excessive category granularity was an issue. The above extraction method produced 23,822 unique categories, raising the possibility that the reported "15% same-category repurchase rate" was an artifact dependent on the degree of category granularity rather than a genuine behavioral pattern.

Fourth, there was no analysis of the substantial data loss during the merging process. Of the 11.2M review records, only 4.5M (approximately 40%) remained after merging with metadata, yet no investigation of the causes or the impact on results was conducted.

Fifth, the scope of analysis was excessively narrow. Only two metrics—category switch rate and brand retention rate—were extracted, while temporal pattern analysis, user behavior classification, and distributional analysis were entirely absent.

Sixth, no statistical validation was performed. The reported "26.39× random" figure was not subjected to any significance testing (p-value, confidence intervals, etc.).

Seventh, the analysis was conducted on a single domain only, making it impossible to determine whether the observed pattern represents a domain-independent, universal phenomenon.

### 1.1.2. Re-conducted EDA

The following improvements were applied to address the above issues.

To ensure consistency between the dataset and the analysis, the title and description were unified to reflect the actual data used (Clothing, Shoes and Jewelry). For category extraction, Level 2 (mid-level categories) was used instead of the lowest level, enabling meaningful product classification while removing noise. Data loss was explicitly documented, including the drop ratio and causes (e.g., 45.3% of items having unregistered brands).

The number of analysis items was expanded from 2 to 5–6, adding category-level repurchase rate distribution, time interval analysis, and user behavior classification. To verify domain independence, the analysis was extended across three independent domains: apparel (Amazon), movies (MovieLens), and books (Goodreads). In the Goodreads analysis, the confounding factor of "reading the next book in the same series" was controlled for, and the hypothesis was confirmed to hold even after this exclusion.

Additionally, the known limitations of each analysis (genre definition noise, single sub-genre verification, ambiguity in timestamp interpretation, etc.) were explicitly stated.

The following publicly available datasets were used for cross-domain validation.

For the apparel domain, the Amazon Clothing, Shoes and Jewelry (5-core, 2018) dataset was used, comprising 6.13M interactions, 1.15M users, and 241K items. The key attribute verified was Brand. (Source: McAuley Lab, UCSD)

For the movie domain, the MovieLens 25M dataset was used, comprising 25.0M ratings, 162.5K users, 54K movies, and 19 primary genres. The key attribute verified was Series (franchise). (Source: GroupLens, UMN)

For the book domain, the Goodreads (children sub-genre) dataset was used, comprising 6.63M interactions, 474K users, and 123K books. The key attribute verified was Author. (Source: Wan & McAuley, UCSD)

## 1.2. Related Work Investigation

After confirming through the re-conducted EDA that the hypothesis holds across multiple domains, we proceeded to investigate existing methodologies that realize the "dynamic graph" direction suggested by our advisor. This ordering was intentional: without data-backed support for the hypothesis, there would be no justification for investigating dynamic knowledge graph-based approaches.

Specifically, the pattern of "brand/author retention during category transitions" observed in the EDA implies that knowledge graph relationships (edge weights, paths, etc.) must be updated whenever a user's purchase event occurs in order to leverage this pattern for recommendation. Accordingly, we surveyed prior work on dynamic knowledge graph-based recommendation, with the following representative studies examined.

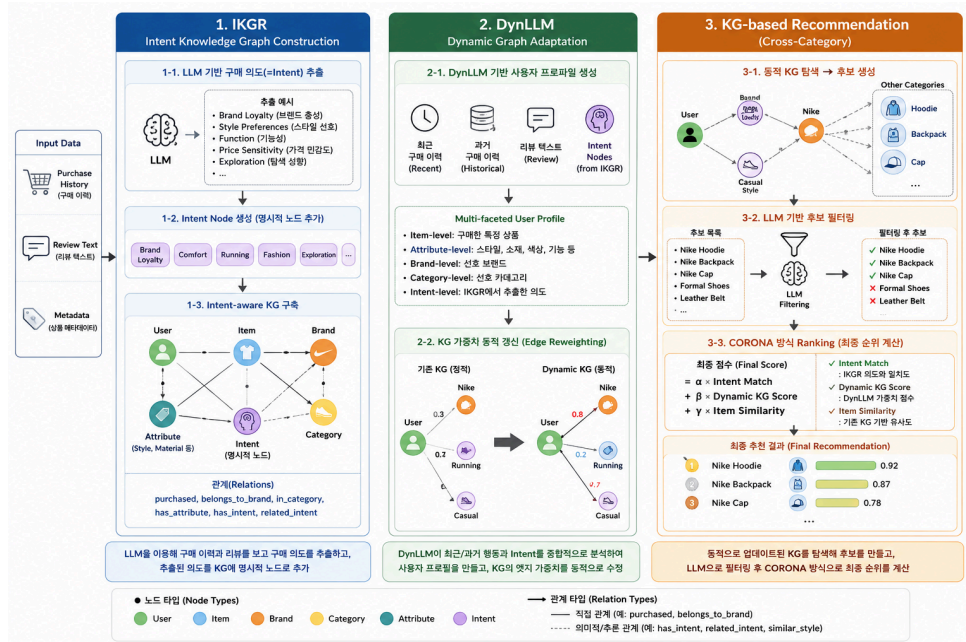
IKGR (Interactive Knowledge Graph Reasoning) is an approach that explores KG paths based on user feedback and explicitly infers purchase intent. DynLLM utilizes large language models (LLMs) to dynamically infer user intent and reflect it in graph updates. CORONA proposes a coarse-to-fine stepwise candidate refinement (ranking) strategy.

## 1.3. Framework Refinement

Based on the related work investigation, we refined and concretized the framework proposed in the Midterm Proposal. The key differences between the original and refined frameworks are as follows.

### 1.3.1 Structural Differences

In the Midterm Proposal, the three modules were designated with generic names (Module 1, 2, 3) and their roles were described conceptually. In the refined framework, the same three-module structure is maintained, but each module is explicitly mapped to a specific prior work. Module 1 is redefined as IKGR-based Intent Knowledge Graph Construction, Module 2 as DynLLM-based Dynamic Graph Adaptation, and Module 3 as CORONA-style KG-based Cross-Category Recommendation.



### 1.3.2 Module 1 (KG Construction) Refinement

In the original design, entities and relations were statically defined with dynamic edge weights applied, but the role of the LLM was concentrated in Module 2, leaving the incorporation of intent in the KG construction phase unclear.

In the refined design, Intent is added as an explicit node in the KG. The LLM analyzes purchase history and review text to extract purchase intents (Brand Loyalty, Comfort, Running, Fashion, Exploration, etc.), generates Intent Nodes, and inserts them into the KG. The relation types now include `has_intent` and `related_intent`, distinguishing between direct relations (`purchased`, `belongs_to_brand`, etc.) and semantic/inferred relations.

The key change is that Intent has been elevated from a "reasoning output" to a "first-class entity (node)" within the KG."

### 1.3.3 Module 2 (Dynamic Graph Update) Refinement

In the original design, the LLM was described as inferring intent types and transition probabilities, which were then used to adjust edge weights—stated conceptually. A GRAM-inspired multi-granular representation was mentioned, but the specific inputs and processing mechanisms were not clearly defined.

In the refined design, DynLLM generates a Multi-faceted User Profile. The inputs include recent purchase history, historical purchase history, review text, and the Intent Nodes generated by Module 1 (IKGR). The user profile is composed of multi-level representations at the Item-level, Attribute-level, Brand-level, and Category-level. Based on this profile, Edge Reweighting is performed on the KG, transforming static weights into dynamic weights, with the process illustrated through numerical examples (e.g., 0.3 → 0.8).

The key change is that the abstract description of "intent inference → edge weight adjustment" has been formalized into a concrete input-processing-output pipeline.

### 1.3.4 Module 3 (Recommendation Generation) Refinement

In the original design, a three-step structure of Coarse Candidate Generation, LLM-guided Candidate Filtering, and Fine-grained Ranking was described, with the final ranking stated to consider intent alignment, KG path scores, and item similarity at a conceptual level.

In the refined design, the same three-step structure is maintained, but the final scoring is explicitly formulated as:  $\text{Final Score} = \alpha \times \text{Intent Match} + \beta \times \text{Dynamic KG Score} + \gamma \times \text{Item Similarity}$ . Each term is mapped to its source module: Intent Match corresponds to alignment with IKGR-inferred intent, Dynamic KG Score corresponds to the DynLLM-updated edge weights, and Item Similarity corresponds to conventional KG-based similarity.

The key change is that the abstract "ranking considers..." has been formalized into a concrete scoring function.

### 1.3.5 Summary of Refinements

The three most significant changes in the refined framework are summarized as follows.

First, explicit connection to prior work. The original framework was presented as if entirely self-designed, whereas the refined version clearly separates the roles of IKGR, DynLLM, and CORONA and restructures the framework as a composition of these approaches.

Second, elevation of Intent's status. Intent, which was previously merely an LLM output (intermediate result), has been elevated to an explicit node within the KG, directly participating in the graph structure.

Third, clarification of inter-module data flow. The original framework left it unclear what information was passed between modules, whereas the refined version explicitly defines the pipeline: Intent Nodes from IKGR serve as input to DynLLM, and edge weights from DynLLM are used for KG traversal in Module 3.

In summary, while the original framework described "what to do" at a conceptual level, the refined framework advances to a concrete design of "how to combine which existing methodologies for implementation."

## 1.4. Current Progress

Our advisor advised that, prior to implementing a new framework design, we should first evaluate how well existing frameworks capture the patterns identified by our proposed hypothesis, and then conduct comparisons. Fortunately, the source code for both IKGR and DynLLM is publicly available. We are currently running the IKGR framework on the Goodreads dataset, which was also used in our EDA, to establish baseline results for comparison.

## 2. Preliminary Results

### 2.1. Results of Re-conducted EDA

The results across the three domains are as follows.<sup>1</sup>

In Amazon Clothing, the category switch rate was 56.4%, and the brand retention rate was 6.53× the random baseline. In MovieLens 25M, the genre switch rate was 70.3%, and the series retention rate was 3.76× the random baseline. In Goodreads (children), the genre switch rate was 63.2%, and the author retention rate was 16.39× the random baseline. The average switch rate across the three domains was 63.3%, and the average attribute retention ratio was 8.89×.

These results consistently support Hypothesis A ("users do not consecutively consume items within the same category"), with switch rates of 56–70% across all domains, and Hypothesis B ("users retain certain attributes even when switching categories"), with retention rates 3.76–16.39× the random baseline. These findings provide sufficient exploratory evidence to propose the limitations of simple category-based recommendation and the necessity of attribute-bridged recommendation via knowledge graphs.

### 2.2. Preliminary Results of IKGR Implementation (Module 1)<sup>2</sup>

Following our advisor's guidance (Section 1.4), the first implementation step was not to build the refined framework from scratch, but to first establish how well an existing intent knowledge graph approach captures the patterns identified in our EDA. Accordingly, we applied IKGR to the Goodreads Children dataset—the same domain used in our cross-domain analysis—to obtain a reliable baseline for subsequent comparison. For model training, we applied a standard  $k=100$  K-core filter, yielding a dense subset of 11,073 users, 6,857 items, and 2,489,355 interactions (approximately 225 interactions per user on average). LLM-based intent extraction (Module 1) produced an intent vocabulary of 53,190 entries, embedded and packed into the knowledge graph as intent nodes (275,554 user–intent edges and 174,535 item–intent edges).

#### Baseline Normalization

The publicly available IKGR implementation did not function as a competitive recommender: its test NDCG@10 was only 0.045, falling below even a non-personalized popularity baseline (Pop, 0.1541). Diagnosis revealed that the scoring function was dominated by a fixed, non-learnable heuristic term whose variance far exceeded the contribution of the learnable component, effectively suppressing the training signal. After redesigning the model around a learnable inner-product score with vectorized intent-node propagation, performance recovered to the 0.29 range, on par with standard collaborative filtering baselines. Crucially, the KG-disabled variant (pure matrix factorization, MF) reached an NDCG@10 of 0.2936, almost identical to BPR (0.2953), serving as a sanity check that confirms the correctness of our implementation.

<sup>1</sup> [https://github.com/HK1DS/graduate\\_eda](https://github.com/HK1DS/graduate_eda)

<sup>2</sup> [https://github.com/HK1DS/IKGR\\_experiment](https://github.com/HK1DS/IKGR_experiment)

*Table 1. Overall comparison after baseline normalization (20 epochs, seed 2020, test set). This table establishes implementation correctness only; it is not directly comparable to Table 2 (different epoch count).*

| Model                  | NDCG@10 | Recall@10 | MRR@10 | Hit@10 |
|------------------------|---------|-----------|--------|--------|
| IKGR (original, buggy) | 0.045   | 0.021     | 0.125  | 0.296  |
| Pop                    | 0.1541  | 0.0765    | 0.3626 | 0.6471 |
| IKGR KG-off (= MF)     | 0.2936  | 0.1526    | 0.5652 | 0.8794 |
| BPR                    | 0.2953  | 0.1537    | 0.5661 | 0.8819 |
| LightGCN               | 0.2964  | 0.1484    | 0.5631 | 0.8671 |
| IKGR KG-on             | 0.2810  | 0.1480    | 0.5469 | 0.8719 |

### From an intent-only KG to a completed heterogeneous KG.

Having established a trustworthy baseline, we evaluated whether enabling the knowledge graph improves recommendation, with particular attention to long-tail items (the bottom 80% of items by popularity) where collaborative-filtering signals are weakest. We report two stages, summarized together in Table 2 under a single multi-seed protocol (12 epochs, seeds 2020/2021/2022) to allow direct comparison.

In the first stage, the KG contained only User, Item, and Intent nodes connected by `has_intent`—that is, only half of our proposed graph design (Section 1.3.2). This intent-only KG did not yield a robust gain: its long-tail Recall@10 (0.0580) was statistically indistinguishable from the MF baseline (0.0597), and its only consistent effect was an increase in recommendation diversity (catalog coverage rising from 0.640 to 0.668). At this stage, the KG’s contribution was best characterized as an accuracy–diversity trade-off rather than an accuracy improvement.

In the second stage, we completed the graph by adding the Brand (author/publisher) and Category/Attribute nodes specified in our framework, with the relations `belongs_to_brand`, `in_category`, and `has_attribute`. These nodes were extracted directly from the Goodreads metadata, requiring no additional LLM calls, yielding a fully deterministic graph of 3,901 author, 744 publisher, and 772 shelf nodes. Once completed in this way, a robust long-tail advantage emerged that the intent-only variant had lacked: the metadata-augmented KG reached a tail Recall@10 of 0.0658 ( $\pm 0.0015$ ), a robust +10.2% improvement over MF and +9.7% over BPR, with low variance. An ablation combining intent and metadata nodes performed almost identically to the metadata-only variant (0.0653 vs 0.0658), indicating that the metadata nodes—particularly category/shelf information—are the primary driver of the gain, while intent adds little on top.

This long-tail advantage comes at the cost of an approximately 8% reduction in overall NDCG@10 (0.268 vs 0.292). We interpret this as the KG improving long-tail exposure and catalog utilization—an explicit objective given the pronounced popularity bias confirmed in strong collaborative filtering (LightGCN attains the highest overall accuracy but the lowest coverage, 0.303)—rather than maximizing overall ranking accuracy.

Table 2. Multi-seed robustness across KG configurations (12 epochs, seeds 2020/2021/2022, mean  $\pm$  std). The intent-only KG does not surpass MF on long-tail accuracy; completing the graph with metadata nodes does, robustly. Bold marks the long-tail leader.

| Model                            | NDCG@10            | tail Recall@10                       | tail Recall@30                       | coverage@10 |
|----------------------------------|--------------------|--------------------------------------|--------------------------------------|-------------|
| KG-off (MF)                      | 0.2922 $\pm$ .0009 | 0.0597 $\pm$ .0008                   | 0.1180 $\pm$ .0010                   | 0.6401      |
| intent-only KG (frozen)          | 0.2782 $\pm$ .0002 | 0.0580 $\pm$ .0002                   | 0.1151 $\pm$ .0004                   | 0.6683      |
| <b>meta-KG (brand/cat/at tr)</b> | 0.2677 $\pm$ .0005 | <b>0.0658 <math>\pm</math> .0015</b> | <b>0.1306 <math>\pm</math> .0029</b> | 0.6566      |
| full hetero (intent + meta)      | 0.2683 $\pm$ .0029 | 0.0653 $\pm$ .0006                   | 0.1292 $\pm$ .0017                   | 0.6500      |
| BPR                              | 0.2935 $\pm$ .0021 | 0.0600 $\pm$ .0015                   | 0.1186 $\pm$ .0013                   | 0.6431      |
| LightGCN                         | 0.2585 $\pm$ .0017 | 0.0276 $\pm$ .0005                   | 0.0566 $\pm$ .0006                   | 0.3028      |

#### Methodological note.

A single-seed run had initially suggested a long-tail advantage even for the intent-only KG (tail Recall@10 of 0.0734 vs 0.0704 for MF), but this did not survive multi-seed validation; we traced the instability to learnable intent nodes, whose variance was an order of magnitude larger than the frozen variant. We report this explicitly because it illustrates the importance of multi-seed validation in preventing overclaims from single-run artifacts—a discipline we maintain throughout. We also note two boundaries of these results. First, the robust long-tail gain is currently established only on the dense  $k=100$  core; verifying cold-start benefits (sparse users) requires a sparser regime, which we address in Section 3. Second, the gain is driven by metadata rather than intent, which directly informs the design of the subsequent modules.

### 3. Plan for Final Submission

We began implementing the pipeline based on the official IKGR source code. In this process, we diagnosed and resolved several deficiencies in the existing implementation—including a scoring function that suppressed the training signal and memory/speed bottlenecks in the evaluation stage—thereby establishing a reliable baseline (Section 2.2). We also observed that training and evaluation incur substantial computational cost even under the relatively dense  $k=100$  setting, an observation we have incorporated into our subsequent data-scale design.

During the summer break, our primary goal is to complete the integrated implementation of the proposed three-module framework. Specifically, building on the now-normalized IKGR (Module 1), we will sequentially integrate DynLLM (Module 2, dynamic edge-weight updating) and CORONA (Module 3, coarse-to-fine candidate refinement with LLM-based filtering and weighted-sum ranking) into a single end-to-end pipeline. Throughout the integration, the normalized IKGR will be reused as a shared component across ablations, allowing the contribution of each module to be measured in isolation.

After completing the framework integration, our final validation step is to lower the data-filtering threshold to a sparse core ( $k=20/30$ ) in order to measure effects under cold-start and long-tail conditions. Since the intent KG was found to contribute to diversity rather than accuracy under the current dense core ( $k=100$ ), contrasting these results with the sparse regime will allow us to address the central question of under which data conditions an intent-aware KG is effective. To this end, we will unify the evaluation protocol (seeds, epochs, metrics) across the dense and sparse experiments to enable direct comparison. Because the transition to a sparse core entails increased cost from a larger volume of LLM-based intent-extraction calls, we will also consider switching to low-cost LLM APIs (e.g., Qwen-turbo, DeepSeek).

## 4. References

### Courses & Lectures

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### Data Sets

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